H-CoReNet: Amplitude-Guided Spectral Filtering and Dictionary-Based Feature Reconstruction for High-Resolution Remote Sensing Segmentation

Yufei Gao, *Member, IEEE*, Zhuangzhi Nie, Liuqian Wang, Yucheng Shi, Dandan Zhang, and Qi Ming

***Abstract*—High-resolution remote sensing (HRRS) image segmentation requires broad semantic context together with accurate boundaries and small-object geometry. H-CoReNet combines three complementary operations for this purpose. The spectral frequency oscillator (SFO) summarizes the magnitude of a 2-D real Fourier transform and uses the resulting descriptor to gate a parallel spatial branch. The harmonic coherence unit (HCU) dynamically weights 3×3, 5×5, and 7×7 depthwise responses and reconstructs the bottleneck feature from an adaptive shared dictionary. The Long-Distance Roaming Window Association (LDRWA) integrates deformable window sampling, mean-centered local modulation, and learnable complex spectral filtering. Experiments use UAVid, an uncrewed aerial vehicle (UAV) urban-scene semantic segmentation benchmark, together with Potsdam, Vaihingen, and LoveDA. Matched test-time augmentation (TTA) and single-scale inference without augmentation (no-TTA) protocols are reported with region- and boundary-sensitive metrics. The quantitative comparison includes recent frequency-domain models, including AFENet and CFormer. Under the matched 512×512 UAVid profiling protocol, the ConvNeXt-Base configuration contains 89.7M parameters and 100.6G FLOPs; the three proposed modules add 0.4M parameters to the backbone-only configuration. The results show that magnitude-guided gating, adaptive dictionary reconstruction, and deformable spectral filtering provide complementary information for remote sensing segmentation. Code is available at https://github.com/ZhuangzhiNie/H-CoReNet**

***Index Terms*—Boundary evaluation, deformable spectral filtering, dictionary reconstruction, high-resolution remote sensing (HRRS), semantic segmentation, spectral magnitude gating.**

NOMENCLATURE

X_{in}, X_{fft}	SFO fused input and real-FFT coefficient tensor.
D_{spec}, ω, X_{spatial}	SFO spectral descriptor, channel weight, and spatial response.
X_{exc}, F_h, F_m, F_l	HCU excited feature and high/middle/low receptive-field responses.
W_h, W_m, W_l	Pixel-wise HCU branch weights.
N_{cls}, K, C	Number of classes, dictionary bases, and feature channels.
B, P_k, R, g_{coh}	Shared dictionary, basis-assignment probabilities, reconstructed tensor, and coherence gate.
Δp	Normalized two-dimensional deformation offset.
X_{win}, Q, K, V	Deformed window tensor and its query, key, and value tensors.
W_{spectral}, V_{recon}	LDRWA complex spectral filter and reconstructed value tensor.
$\odot$	Element-wise product.

I. INTRODUCTION

SEMANTIC segmentation of high-resolution remote sensing (HRRS) images is essential for Earth observation. It supports fine land-cover mapping applications [13], [51]. Complex urban morphologies, illumination, and haze effects increase scene parsing difficulty. Remote sensing targets commonly exhibit interclass homogeneity and intraclass heterogeneity [70]. These characteristics require stable feature representation capabilities from segmentation models. Accurately delineating boundaries of complex tiny geographic objects remains a research focus [64].

Mainstream vision models often rely on traditional encoder–decoder architectures [49], [51]. These architectures fuse multiscale features through layerwise cascading. This unidirectional serial transmission causes shallow geometric information loss. Deep networks extract high-level semantics containing global context. Macroscopic semantics often face information dilution during progressive upsampling. Limited receptive fields lead to structural misjudgments when models process smooth regions [48]. Spatial-domain feature fusion mechanisms present inherent limitations.

From a signal-processing perspective, homogeneous regions concentrate more energy at low spatial frequencies, whereas abrupt boundaries contribute relatively more high-frequency energy. A convolutional feature channel may contain both components, and neural networks can exhibit spectral bias [57]. H-CoReNet therefore combines spectral descriptors,

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learned spectral filters, and spatial receptive-field branches to represent these complementary responses.

Based on this analysis, H-CoReNet combines three complementary modules. The spectral frequency oscillator (SFO) performs spectral-magnitude gating, whereas the harmonic coherence unit (HCU) couples adaptive receptive-field mixing with dictionary reconstruction. The Long-Distance Roaming Window Association (LDRWA) combines deformable sampling with complex spectral filtering. The aggregation head aligns the four decoder scales before prediction. Experiments include UAVid, an uncrewed aerial vehicle (UAV) urban-scene semantic segmentation benchmark, alongside Potsdam, Vaihingen, and LoveDA.

The contributions are summarized in four aspects.

1) We construct H-CoReNet as an HRRS segmentation framework that combines direct multistage feature aggregation with amplitude-guided filtering and dictionary-based feature reconstruction. The complete ConvNeXt-Base model contains approximately 89.7M parameters, and the added modules introduce controlled overhead.
2) We design SFO as an amplitude-only spectral gating module. It uses a 2-D real Fourier transform to extract channel-level magnitude descriptors, while a parallel spatial branch and residual connection preserve localization information.
3) *We introduce HCU as a coupled bottleneck operator:* pixelwise mixing of 3×3, 5×5, and 7×7 depthwise responses, condition assignment to a shared class-agnostic dictionary, and the dictionary reconstruction is injected through a coherence gate. This coupling differs from convolution-only multiscale attention and from iterative class-embedding dictionary refinement. Its shared dictionary size is $K = \max(24, \min(8N_{\text{cls}}, 128))$, yielding 64, 48, 48, and 56 bases for UAVid, Potsdam, Vaihingen, and LoveDA, respectively.
4) We integrate LDRWA to capture nonrigid topologies through deformable local-window aggregation. We evaluate the method under test-time augmentation (TTA) and single-scale inference without augmentation (no-TTA), together with boundary-sensitive metrics, robustness analysis, and computational-efficiency evidence.

II. Related Work

A. CNN-Based Remote Sensing Image Parsing

Convolutional neural networks (CNNs) established the foundation for remote sensing image parsing. FCN introduced the end-to-end pixel-level classification paradigm [30]. U-Net relies on an encoder–decoder architecture to progressively restore spatial dimensions [41]. Multiscale feature fusion expands receptive fields. For example, PSPNet extracts hierarchical signals through spatial pyramid pooling [63]. Standard convolutional kernels capture local short-range visual features. Single pooling dimensionality reduction triggers homogenized information redundancy. High-level semantics attenuate as networks deepen. Cross-boundaries of complex urban landscapes blur during cascaded transmission [65]. Local geometric features limit high-precision full-element mapping. Optimizing underlying signal extraction mechanisms addresses these limitations.

B. Global Spatial Association and ViTs

Local convolutions possess limited capacity to capture macroscopic global dependencies. Pixel-level interaction mechanisms establish long-distance perception capabilities. DANet synchronously constructs spatial and channel dual attention flows [9]. Vision Transformers (ViTs) discretize images into 1-D sequences for processing [7]. The Swin architecture utilizes sliding windows to reduce spatial computational loads [28]. Global pixel-level dot products introduce computational redundancy. Processing HRRS images generates high memory overhead [48]. Sequence decoders often discard high-frequency geometric contour details. An LDRWA mechanism addresses these structural issues. This mechanism enables macroscopic spatial and nonrigid topological perception [55]. It simultaneously controls computational overhead.

C. Frequency-Domain Feature Modeling for Remote Sensing Segmentation

Frequency-domain processing has been used to separate or modulate spatial-frequency responses in dense prediction. Representative designs include octave convolution, frequency channel attention, global frequency filtering, Fourier-domain adaptation, and boundary-oriented spectral modulation [2], [3], [38], [40], [58]. Recent remote sensing segmentation networks further integrate adaptive frequency enhancement, frequency–spatial interaction, and frequency-guided decoding [8], [10], [14], [27], [62], [67], [68]. PyramidMamba provides a state-space reference for long-range multiscale aggregation [48].

AFENet uses adaptive high- and low-frequency masks with selective fusion for remote sensing semantic segmentation [10]. CFormer combines DCT-based frequency-aware channel attention with a Fourier relational transformer [53]. IGECNet combines HRRSI and DSM inputs with an FFT-guided context decoder, local boundary refinement, and elevation constraints [59]. SegNeXt provides a convolutional attention reference for semantic segmentation [12]. Three additional TGRS studies extend this context to frequency-domain change detection, frequency-guided denoising for semantic segmentation, and multimodal global–local frequency reconstruction [26], [43], [54].

FSSFFormer combines fixed wavelet decomposition with spectral enhancement and spectral–spatial context attention to preserve global structure and fine detail in remote sensing segmentation [23]. FDNet refines high- and low-frequency components separately to improve feature representations for remote sensing image segmentation [25]. GPINet couples CNN and transformer features through local–global interaction modules and uses learned geometric priors to guide feature recovery during decoding [24].

Mechanism-level comparison places HCU between multiscale context modeling and prototype reconstruction. SegNeXt

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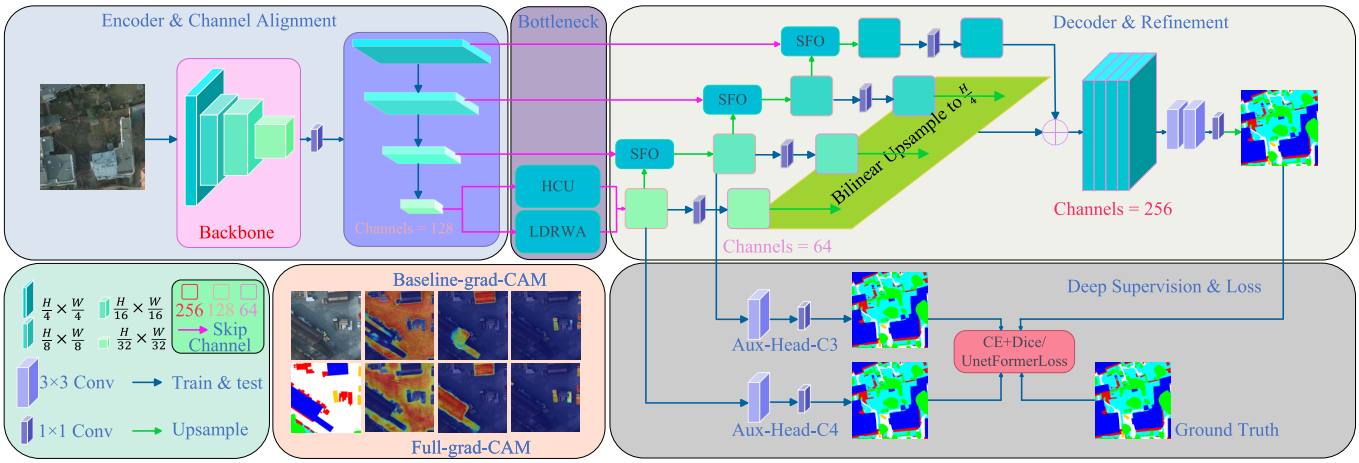


Fig. 1. Overall architecture of H-CoReNet. HCU and LDRWA process the same deepest aligned feature $\bar{c}_4$ in parallel, and their outputs are added to form d_4. Three SFO blocks successively combine d_4, d_3, and d_2 with the encoder features $\bar{c}_3$, $\bar{c}_2$, and $\bar{c}_1$, respectively. The aggregation head aligns and concatenates d_1–d_4 before prediction. Auxiliary heads provide deep supervision only during training.

encodes multiscale spatial context with convolutional attention but has no explicit dictionary-reconstruction stage [12]. ScaleProtoSeg learns class-linked prototypical parts at multiple decoder scales and sparsely groups their activations for interpretable segmentation decisions [37]. D2LS constructs class-aware dictionary embeddings and iteratively refines them through alternating cross-attention with image features and a contrastive dictionary constraint [70]. HCU instead uses the pixelwise mixture of three depthwise receptive-field responses to predict one locationwise assignment to a shared class-agnostic dictionary, followed by one-pass reconstruction and coherence-gated residual injection at the bottleneck. Its contribution is therefore the conditional coupling of scale selection and dictionary-constrained reconstruction, rather than multiscale convolution or prototype learning in isolation.

Four 2026 top-venue studies sharpen this distinction. PMDAv2 aggregates confidence-weighted multiscale class prototypes for source–target alignment in domain-adaptive segmentation [22]. Visual description assembly dynamically composes instance-specific visual prototypes as CLIP queries for weakly supervised segmentation [39]. HyperProtoSeg learns class prototypes in a Poincaré ball and adapts them with boundary-aware consistency under continual test-time shift [11]. DCMIF is the closest recent shared-dictionary reconstruction reference: it learns convolutional atoms, infers modality coefficients, and reconstructs a fused image in a missing-infrared setting [61]. These methods use prototypes for domain alignment, query construction, online adaptation, or cross-modal image reconstruction. HCU instead predicts spatial assignments from a mixed bottleneck feature to a shared $K \times 128$ feature dictionary and injects $B^{\top}\widetilde{P}$ into the same segmentation stream. This comparison limits the novelty claim to the implemented within-sample sequence and does not treat "harmonic coordination" as physical resonance.

H-CoReNet places magnitude-guided channel gating in decoder transmission, adaptive multireceptive-field weighting, and dictionary reconstruction at the bottleneck, and complex spectral filtering within deformable local windows. This arrangement links global structure, local geometry, and shared semantic prototypes in a single encoder–decoder pipeline. Related spatial designs based on boundary alignment, deformable sampling, and high-resolution feature preservation provide complementary context [15], [46], [60], [69].

III. Methodology

A. Overall Architecture

H-CoReNet combines bottleneck refinement with progressive decoder transmission (Fig. 1). HCU and LDRWA operate in parallel on the deepest aligned feature. HCU provides adaptive receptive-field mixing and shared-dictionary reconstruction, while LDRWA provides deformable local modulation and complex spectral filtering. SFO subsequently transfers their combined representation through the decoder using spectral-magnitude channel gates.

In this article, *feature aliasing* denotes boundary leakage or interclass confusion associated with mixed activations. *Harmonic coordination* denotes the joint modulation produced by HCU branch weights and shared dictionary responses. The *aggregation field* denotes the aligned multiscale tensor supplied to the prediction head.

The backbone produces four hierarchical features, c_1–c_4, which are projected to a common channel dimension to obtain $\bar{c}_1$–$\bar{c}_4$. The refined bottleneck and three decoder stages are

$$\begin{aligned} d_4 &= \mathrm{HCU}\left(\bar{c}_4\right) + \mathrm{LDRWA}\left(\bar{c}_4\right) \\ d_i &= \mathrm{SFO}\left(d_{i+1}, \bar{c}_i\right), \quad i = 3, 2, 1. \end{aligned} \tag{1}$$

Thus, HCU and LDRWA receive the same input, and neither branch takes the other branch's output as its input. LDRWA supplies deformable spatial support [52], while HCU supplies dictionary-constrained context. Their sum becomes the high-level input of the first SFO block. Each subsequent SFO block combines the preceding decoder output with the corresponding aligned encoder feature, following (3)–(6). The prediction head aggregates all four decoder scales as

$$\widehat{Y} = \mathcal{H}_{\mathrm{pred}}\left(\mathrm{Concat}_{i=1}^{4}\mathcal{A}_i\left(d_i\right)\right) \tag{2}$$

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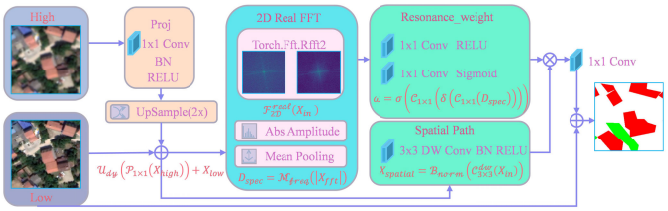


Fig. 2. SFO aligns high- and low-level features, computes the spectral descriptor D_{spec}, generates the channel weight ω, and gates the parallel response X_{spatial}.

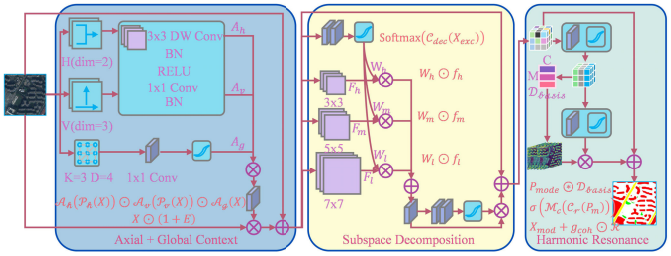


Fig. 3. HCU forms X_{exc} from the axial and global context, weights the F_h, F_m, and F_l branches, and reconstructs R from P_k and the shared dictionary$B$.

where $\mathcal{A}_i$ projects and resizes each decoder tensor to the common aggregation grid and $\mathcal{H}_{\text{pred}}$ produces the semantic prediction.

The auxiliary heads provide deep supervision during training and are removed at inference. Cross-entropy and Dice losses are applied to the main and auxiliary outputs.

B. Spectral Frequency Oscillator

SFO transfers decoder information using a spectral descriptor and a parallel spatial path. Given X_{high} and X_{low}, a 1×1 projection and dynamic upsampling align the two tensors

$$X_{\text{in}} = \mathcal{U}_{\text{dy}}\left(\mathcal{P}_{1\times1}\left(X_{\text{high}}\right)\right) + X_{\text{low}}. \tag{3}$$

The real 2-D Fourier transform uses orthogonal normalization. Mean pooling of its magnitude produces the descriptor shown in Fig. 2

$$\begin{aligned} X_{\text{fft}} &= \mathcal{F}_{2D}^{\text{real}}\left(X_{\text{in}}; \text{ortho}\right) \\ D_{\text{spec}} &= \mathcal{M}_{\text{freq}}\left(|X_{\text{fft}}|\right). \end{aligned} \tag{4}$$

Two pointwise convolutions generate the channel weight

$$\omega = \sigma\left(\mathcal{C}_{1\times1}^{(2)}\left(\delta\left(\mathcal{C}_{1\times1}^{(1)}\left(D_{\text{spec}}\right)\right)\right)\right). \tag{5}$$

The spatial path applies depthwise convolution, batch normalization, and ReLU. The weighted response is projected and added to the low-level residual

$$\begin{aligned} X_{\text{spatial}} &= \mathcal{B}_{\text{norm}}\left(\mathcal{C}_{3\times3}^{\text{dw}}\left(X_{\text{in}}\right)\right) \\ Y_{\text{SFO}} &= \mathcal{C}_{1\times1}\left(X_{\text{spatial}} \odot \omega\right) + X_{\text{low}} \end{aligned} \tag{6}$$

where $\mathcal{B}_{\text{norm}}$ denotes batch normalization followed by ReLU. The descriptor D_{spec} controls channel selection, while X_{spatial} and the residual X_{low} preserve spatial localization.

C. Harmonic Coherence Unit

HCU combines axial context, three depthwise receptive-field branches, and dictionary-based reconstruction. Its structural distinction is the ordering and conditioning of these operations. Unlike a convolution-only multiscale attention block, HCU uses the mixed receptive-field feature to predict spatial dictionary assignments. Unlike D2LS, which iteratively updates class-aware dictionary embeddings through alternating cross-attention [70], HCU learns one shared class-agnostic dictionary and performs a single locationwise assignment and gated reconstruction at the bottleneck. Horizontal pooling $\mathcal{P}_h$, vertical pooling $\mathcal{P}_v$, and the dilated global path $\mathcal{A}_g$ form the excitation tensor

$$\begin{aligned} E &= \mathcal{C}_{1\times1}\left(\mathcal{A}_h\left(\mathcal{P}_h\left(X\right)\right) \odot \mathcal{A}_v\left(\mathcal{P}_v\left(X\right)\right) \odot \mathcal{A}_g\left(X\right)\right) \\ X_{\text{exc}} &= X \odot (1 + E). \end{aligned} \tag{7}$$

The subspace decomposer predicts the three branch weights

$$\begin{aligned} [W_h, W_m, W_l] &= \text{Softmax}(\mathcal{C}_{\text{dec}}\left(X_{\text{exc}}\right)) \\ F_b &= \mathcal{C}_{k_b\times k_b}^{\text{dw}}\left(X_{\text{exc}}\right), \quad b \in \{h, m, l\} \\ (k_h, k_m, k_l) &= (3, 5, 7). \end{aligned} \tag{8}$$

The weighted branch responses are concatenated and projected before residual modulation

$$\begin{aligned} F_{\text{band}} &= \mathcal{C}_{1\times1}\left(\text{Concat}\left(W_h \odot F_h, W_m \odot F_m, W_l \odot F_l\right)\right) \\ X_{\text{mod}} &= X_{\text{exc}} + \sigma\left(\mathcal{M}\left(F_{\text{band}}\right)\right) \odot F_{\text{band}}. \end{aligned} \tag{9}$$

For branch b, the convolution theorem gives $\widehat{F}_b(u, v) = \widehat{H}_b(u, v)\widehat{X}_{\text{exc}}(u, v)$. The learned kernels therefore modulate spatial-frequency responses, while the three HCU branches remain spatial receptive-field operators.

The dictionary in Fig. 3 contains K shared learnable bases with $C = 128$ channels

$$\begin{aligned} B &\in \mathbb{R}^{K\times128} \\ K &= \max(24, \min\left(8N_{\text{cls}}, 128\right)) \end{aligned} \tag{10}$$

where N_{cls} denotes the number of semantic classes and scales the size of one shared dictionary. The factor eight does not assign eight bases to each class. The lower and upper bounds retain a minimum shared capacity and cap its growth with class count. The rule yields 64 bases for UAVid, 48 for Potsdam, 48 for Vaihingen, and 56 for LoveDA. We use it as a practical capacity–cost compromise, supported by the broad Potsdam performance plateau around $K = 48$. This evidence does not establish a statistically optimal dictionary size for every dataset. Each additional basis contributes 128 dictionary parameters and 385 parameters to the complete K-dependent HCU path.

A 1×1 projection and channelwise softmax generate the basis-assignment probabilities P_k

$$(P_k)_{b,k,h,w} = \frac{\exp\left(L_{b,k,h,w}\right)}{\sum_{q=1}^{K}\exp\left(L_{b,q,h,w}\right)}. \tag{11}$$

The assignment probabilities combine the shared bases into the reconstructed tensor R

$$R_{b,c,h,w} = \sum_{k=1}^{K}(P_k)_{b,k,h,w}\, B_{k,c}. \tag{12}$$

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After flattening the spatial dimensions to $N = HW$, the reconstruction can be written as

$$\widetilde{R} = B^{\mathrm{T}}\widetilde{P}, \quad \operatorname{rank}\left(\widetilde{R}\right) \le K \tag{13}$$

where $\widetilde{P} \in \mathbb{R}^{K\times N}$ and $\widetilde{R} \in \mathbb{R}^{C\times N}$. This factorization constrains the reconstructed response to the span of the K learned prototypes while allowing the assignment vector to vary spatially. Increasing K raises the rank bound and increases the assignment and reconstruction costs approximately linearly. The coherence gate and HCU output follow the notation in Fig. 3:

$$\begin{aligned} g_{\mathrm{coh}} &= \sigma\left(\mathcal{M}_c\left(\mathcal{C}_r\left(P_k\right)\right)\right) \\ Y_{\mathrm{HCU}} &= X_{\mathrm{mod}} + g_{\mathrm{coh}} \odot R \end{aligned} \tag{14}$$

where *harmonic coordination* names the ordered coupling of pixelwise receptive-field mixing, shared-dictionary assignment, and gated residual reconstruction. The dictionary bases are learned feature prototypes rather than Fourier harmonic functions; the term *harmonic* refers to coordination among these responses.

The novelty claim is therefore restricted to the coupled operator and its placement at the shared bottleneck. The ablation study isolates the HCU-only contribution across four datasets, the sensitivity analysis evaluates the capacity parameter K, and the basis visualization shows the spatial use of all learned bases. These analyses support the contribution of adaptive branch mixing and prototype reconstruction without treating the constituent convolutions or dictionary lookup as new primitives. A mechanism-level comparison with SegNeXt and D2LS is provided in the Supplementary Material section "Structural Positioning of HCU."

D. Long-Distance Roaming Window Association

LDRWA combines deformable sampling, mean-centered local modulation, and complex spectral filtering (Fig. 4). The offset network contains a dilated 5×5 depthwise convolution, batch normalization, GELU, and a 1×1 projection. It predicts Δp, and the warped feature is partitioned into 7×7 windows

$$\begin{aligned} \Delta p &= 0.5 \tanh\left(\mathcal{C}_{\mathrm{off}}\left(X\right)\right) \\ X_{\mathrm{win}} &= \mathcal{S}_{\mathrm{window}}\left(\mathcal{W}_{\mathrm{grid}}\left(X, \Delta p\right)\right). \end{aligned} \tag{15}$$

The pointwise QKV projection is divided into four heads. With $s = d_h^{-1/2}$, the response tensor is

$$\begin{aligned} [Q, K, V] &= \mathrm{Chunk}\left(\mathcal{C}_{qkv}\left(X_{\mathrm{win}}\right)\right) \\ R &= \sigma\left(\left(Q - \bar{Q}\right) \odot \left(K - \bar{K}\right)\, s\right). \end{aligned} \tag{16}$$

The value tensor is filtered by the learnable complex weight W_{spectral}

$$V_{\mathrm{recon}} = \mathcal{F}_{2D}^{-1}\left(\mathcal{F}_{2D}^{\mathrm{real}}\left(V\right) \otimes W_{\mathrm{spectral}}\right) \tag{17}$$

where $\otimes$ denotes elementwise complex multiplication. The response R modulates V_{recon} before window merging and inverse warping. A parallel 3×3 depthwise path supplies F_l

$$\begin{aligned} F_g &= \mathcal{C}_{1\times 1}\left(\mathcal{W}_{\mathrm{grid}}^{-1}\left(R \odot V_{\mathrm{recon}}, -\Delta p\right)\right) \\ Y_{\mathrm{LDRWA}} &= X + (1-\alpha)\, F_g + \alpha F_l. \end{aligned} \tag{18}$$

The learnable parameter α balances the inverse-warped spectral branch and the local spatial branch.

IV. Experiments

A. Datasets

Experiments evaluate model parsing capabilities on four remote sensing semantic segmentation datasets. The datasets include the UAV urban-scene benchmark UAVid [34], the aerial benchmarks Potsdam [17] and Vaihingen [18], and the satellite benchmark LoveDA [47]. Data partitioning and preprocessing follow standard protocols [16], [51], [70]. LoveDA and UAVid serve online testing scenarios with undisclosed labels. Appendix A provides detailed parameters and native resolutions for each dataset.

B. Implementation Details

Experiments run on a single NVIDIA RTX 4090 GPU using PyTorch. AdamW uses a weight decay of 0.01, and a cosine schedule controls the learning rate [31], [32]. ConvNeXt-Base and EfficientNet-B3 are initialized from ImageNet weights [29], [44]. The initial learning rate is 1×10^{-3}; the newly introduced layers use 1×10^{-4} with ConvNeXt-Base and 9×10^{-3} with EfficientNet-B3.

The training schedule is specified for each dataset and backbone. On UAVid, all backbones are trained for 15 epochs except EfficientNet-B3, which is trained for 45 epochs to obtain a converged checkpoint. All LoveDA backbones are trained for 15 epochs. Potsdam is trained for 45 epochs, whereas Vaihingen is trained for 105 epochs. UAVid uses 1024×1024 crops with batch size 2; the other datasets use 512×512 crops with batch size 8. Repeated-run experiments use seeds $\{0, 42, 84, 126, 168\}$.

Training augmentation includes random flipping and 90° rotations. Potsdam, Vaihingen, and LoveDA additionally use random scaling from 0.5 to 1.5. Color jitter, Gaussian blur, and coarse dropout are applied to Potsdam, UAVid, and LoveDA, while Vaihingen uses geometric augmentation only to preserve IRRG radiometry.

All methods reported in Tables I–VIII were retrained from their released or official implementations using the same dataset split and five seeds $\{0, 42, 84, 126, 168\}$. TTA combines predictions from scaled and flipped inputs. Within each dataset, every method uses the same inference transforms. UAVid uses scales $\{0.75, 1.0, 1.25, 1.5\}$ with horizontal flipping; Potsdam and Vaihingen use the same four scales with horizontal and vertical flipping; LoveDA uses scales $\{0.5, 0.75, 1.0, 1.25, 1.5\}$ with horizontal and vertical flipping. Bicubic interpolation is used with `align_corners=False`. TTA is applied to each evaluation input without an additional center crop; each prediction is inverse-transformed to the original evaluation-input size, and the deaugmented logits are fused by an arithmetic mean before the classwise argmax. The no-TTA protocol uses only the 1.0-scale input without flipping. The main article reports the five-seed TTA results, and the Supplementary Material reports no-TTA results from the corresponding checkpoints.

C. Evaluation Metrics

Region-level performance is measured using overall accuracy (OA), mean intersection over union (mIoU), and mean

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TABLE I

QUANTITATIVE COMPARISON ON THE UAVID TEST SET UNDER THE TTA PROTOCOL

Method	Backbone	Clutter	Building	Road	Tree	Vegetation	MovingCar	StaticCar	Human	mIoU↑	mB-F1	mTri-IoU	mB-IoU	mS-IoU
DANet [9]	ResNet18	66.54±0.52	86.03±0.44	80.51±0.60	80.17±0.13	63.33±0.59	71.47±0.91	65.54±2.56	25.26±0.54	67.36±0.33	48.14±0.22	50.89±0.28	18.67±0.09	41.47±0.51
ABCNet [21]	ResNet18	67.28±0.42	86.51±0.36	81.36±0.29	80.07±0.21	63.42±0.34	70.16±0.67	49.07±1.43	14.21±0.48	64.01±0.21	46.49±0.29	49.52±0.24	17.61±0.08	40.08±0.68
BANet [50]	ResT-Lite	66.84±0.38	85.62±0.31	80.93±0.27	79.14±0.23	62.37±0.32	69.58±0.59	53.26±1.18	21.34±0.41	64.89±0.24	46.93±0.27	49.89±0.21	18.04±0.09	40.57±0.61
Segmenter [56]	Vit-Tiny	65.91±0.31	85.84±0.24	79.48±0.29	79.11±0.18	61.72±0.37	71.68±0.63	50.94±1.42	27.13±0.46	65.23±0.28	47.71±0.25	50.61±0.23	18.43±0.07	41.08±0.64
UNetFormer [51]	ResNet18	66.81±0.20	86.67±0.32	80.98±0.23	80.05±0.08	63.62±0.29	73.36±0.49	57.21±1.17	30.67±0.43	67.42±0.20	50.47±0.14	52.41±0.11	19.94±0.05	42.71±0.57
SFA-Net [16]	EfficientNet-B3	69.31±0.33	88.45±0.30	82.75±0.23	80.72±0.24	64.11±0.35	**77.02±0.22**	64.57±2.42	30.37±0.28	69.66±0.41	55.63±0.18	56.97±0.16	22.91±0.07	47.18±0.36
D2LS [70]	ConvNeXt-Base	70.65±0.14	89.14±0.11	82.24±0.17	81.71±0.09	**66.26±0.19**	73.74±0.75	61.48±1.21	**33.00±0.24**	69.78±0.18	55.46±0.13	56.74±0.12	22.63±0.04	48.03±0.24
AFENet [10]	ResNet18	69.16±0.55	88.48±0.20	82.66±0.55	80.61±0.15	64.37±0.15	75.01±1.46	62.55±2.04	32.00±0.13	69.36±0.28	59.98±0.12	59.46±0.06	24.52±0.08	49.25±0.62
CFormer [53]	ResNet50	63.97±0.18	84.72±0.83	79.39±1.29	78.36±0.68	61.45±0.08	57.32±18.28	48.40±13.06	27.49±0.03	62.64±0.62	46.51±0.25	51.24±0.40	17.90±0.14	39.20±1.03
H-CoReNet (Ours)	EfficientNet-B3	68.95±0.54	88.30±0.31	**82.92±0.37**	80.69±0.09	64.43±0.17	76.17±1.31	64.61±2.61	31.34±0.42	69.68±0.41	58.24±0.11	58.08±0.13	23.73±0.09	48.34±0.52
H-CoReNet (Ours)	ConvNeXt-Base	**71.13±0.24**	**89.72±0.19**	82.69±0.26	**81.82±0.17**	66.13±0.31	76.08±0.62	**72.43±1.27**	32.29±0.28	**71.54±0.17**	**60.42±0.14**	**59.86±0.16**	**25.07±0.09**	**51.14±0.57**

TABLE II

CLASSWISE BOUNDARY-SENSITIVE COMPARISON ON UAVID UNDER THE TTA PROTOCOL. B-F1, TRI-IOU, B-IOU, AND S-IOU ARE COMPUTED WITH THE SAME CLASS MASKS AND BOUNDARY TOLERANCE

Method	Building				Road				MovingCar				StaticCar			
	B-F1	Tri-IoU	B-IoU	S-IoU	B-F1	Tri-IoU	B-IoU	S-IoU	B-F1	Tri-IoU	B-IoU	S-IoU	B-F1	Tri-IoU	B-IoU	S-IoU
DANet [9]	41.73±0.46	53.84±0.59	15.58±0.21	31.18±1.34	52.16±0.27	55.93±0.49	20.87±0.14	32.05±1.96	76.92±0.43	67.31±0.54	36.37±0.29	73.04±1.29	53.67±0.51	54.83±0.61	21.02±0.23	48.56±1.21
ABCNet [21]	40.08±0.41	53.11±0.54	14.76±0.17	32.03±1.61	50.62±0.34	54.86±0.46	19.71±0.13	31.44±1.47	75.46±0.63	66.57±0.58	35.28±0.37	72.01±1.42	50.93±0.57	53.62±0.69	19.21±0.26	47.38±1.63
BANet [50]	40.59±0.37	53.39±0.51	15.11±0.15	32.36±1.39	51.14±0.31	55.31±0.42	20.12±0.11	31.84±1.26	76.09±0.54	66.94±0.51	35.73±0.31	72.68±1.28	51.82±0.49	54.07±0.63	19.68±0.21	47.83±1.49
Segmenter [56]	41.97±0.29	53.96±0.43	15.86±0.12	32.91±1.16	51.84±0.25	55.82±0.38	20.64±0.09	32.69±1.13	77.61±0.39	67.86±0.44	36.98±0.24	74.22±1.08	53.51±0.42	55.39±0.56	20.93±0.18	48.87±1.26
UNetFormer [51]	44.53±0.16	55.36±0.34	17.14±0.07	30.27±1.51	54.68±0.19	57.61±0.31	22.34±0.10	34.83±1.17	79.21±0.27	69.18±0.24	38.47±0.16	75.81±0.61	56.42±0.31	57.03±0.46	22.39±0.14	50.39±1.42
SFA-Net [16]	47.79±0.27	56.48±0.24	18.34±0.12	31.06±1.03	59.11±0.23	60.41±0.27	23.76±0.14	39.32±1.31	82.37±0.19	70.72±0.21	39.83±0.13	80.06±0.44	65.19±0.36	62.11±0.51	26.31±0.19	59.82±1.56
D2LS [70]	47.14±0.19	56.03±0.17	18.17±0.08	34.52±0.91	58.31±0.18	59.84±0.21	23.39±0.07	39.92±0.63	81.06±0.15	70.83±0.11	38.99±0.09	81.04±0.32	63.34±0.25	61.52±0.29	25.27±0.10	59.49±0.79
AFENet [10]	51.93±0.20	58.23±0.11	19.83±0.08	25.05±1.73	**65.20±0.24**	62.47±0.21	27.00±0.14	42.48±1.47	84.99±0.21	71.64±0.23	40.68±0.23	82.43±0.53	74.67±0.36	**67.54±0.08**	31.72±0.26	**69.14±1.34**
CFormer [53]	35.77±0.18	54.97±0.93	12.75±0.01	33.12±10.23	52.13±0.30	55.09±0.16	20.19±0.11	32.72±5.19	73.81±1.56	62.20±1.87	33.12±0.57	64.22±3.80	59.49±1.09	57.89±0.89	23.49±0.83	44.58±3.08
H-CoReNet (Ours)	50.74±0.18	57.83±0.39	19.20±0.13	31.22±1.65	62.98±0.19	61.41±0.29	25.82±0.11	41.67±1.88	84.53±0.22	71.23±0.22	40.77±0.18	81.19±0.65	73.66±0.35	65.00±0.29	31.43±0.27	63.05±1.47
H-CoReNet (Ours)	**52.11±0.23**	**59.46±0.36**	**20.24±0.12**	**35.83±1.21**	65.07±0.26	**63.14±0.32**	**27.36±0.14**	**44.21±1.49**	**86.02±0.24**	**72.61±0.29**	**41.96±0.18**	**83.17±0.61**	**75.16±0.39**	66.93±0.44	**32.58±0.19**	66.07±1.14

$F1$ score (mF1) following established remote-sensing segmentation practice [33]. Boundary behavior is measured using boundary $F1$ (B-F1), trimap IoU (Tri-IoU), boundary IoU (B-IoU), and small-structure IoU (S-IoU) [5], [19]. For every test image and class, one-pixel inner contours are extracted from the binary prediction and ground-truth masks. B-F1 and B-IoU use an elliptical dilation radius of $r_b = 3$ pixels, and Tri-IoU uses a ground-truth-boundary trimap radius of $r_t = 5$ pixels. For S-IoU, eight-neighbor connected components are extracted separately from each image–class ground-truth mask, and components satisfying $A \leq A_s = 1024$ pixels define the small-structure region. The evaluator accumulates the per-class boundary matches, boundary pixels, intersections, and unions across all test images within a dataset and computes each ratio only after this accumulation; the reported values are therefore dataset-level ratios, not averages of per-image scores. Dataset means are arithmetic means over classes with valid denominators. UAVid and LoveDA include all labeled classes, whereas Potsdam and Vaihingen follow the standard ISPRS protocol and exclude the clutter class from dataset-level means. The same implementation is used for every model and for TTA and no-TTA.

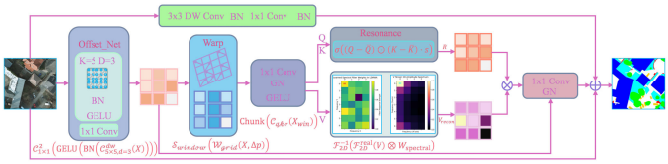


Fig. 4. LDRWA predicts Δp, forms X_{win}, computes Q, K, V, and R, filters V with W_{spectral}, and fuses the reconstructed global response with the local branch.

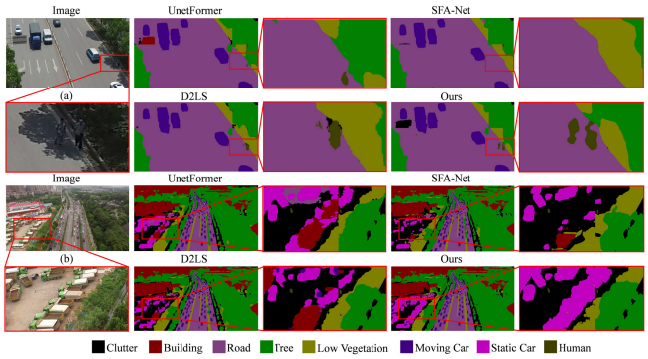


Fig. 5. Native-resolution qualitative comparison on UAVid frames of up to 4096 × 2160 pixels. Selected regions and magnified views retain the large-scene context. (a) H-CoReNet localizes tiny pedestrians under illumination and haze effects and outlines traffic-sign contours. (b) Model separates dense static vehicles under complex background interference.

D. Comparative Experiments and Analysis

1) UAVid Dataset Experimental Results: The UAVid dataset contains complex urban street-view panoramas with native frames up to 4096 × 2160 pixels. UAV oblique photography causes spatial scale variations [66]. This characteristic increases remote sensing object segmentation difficulty. Experiments evaluate H-CoReNet against existing architectures on this benchmark.

Table I reports the unified TTA region-and-boundary comparison, and Table II reports classwise boundary metrics. On UAVid, EfficientNet-B3 uses the converged 45-epoch checkpoint, while all other backbones retain the 15-epoch schedule specified in the implementation details. H-CoReNet with ConvNeXt-Base achieves 71.54% ±0.17% mIoU, compared with 69.78% ±0.18% for D2LS, and records 60.42% ±0.14% mB-F1, 59.86% ±0.16% mTri-IoU, 25.07% ±0.09% mB-IoU, and 51.14% ±0.57% mS-IoU. The EfficientNet-B3 configuration reaches 69.68% ±0.41% mIoU after the 45-epoch training schedule.

Fig. 5 compares the output masks produced by the evaluated models. In Fig. 5(a), illumination and haze obscure the tiny

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TABLE III

QUANTITATIVE COMPARISON ON THE POTSDAM TEST SET UNDER THE TTA PROTOCOL

Method	Backbone	Imp.surf.	Building	LowVeg	Tree	Car	mF1	mB-F1	mTri-IoU	mB-IoU	mS-IoU
DANet [9]	ResNet18	91.36±0.12	95.82±0.11	86.35±0.14	87.89±0.10	84.63±0.06	89.21±0.08	66.52±0.15	80.85±0.21	51.92±0.11	56.88±0.32
ABCNet [21]	ResNet18	92.93±0.15	96.24±0.11	87.15±0.21	88.34±0.14	95.27±0.08	91.99±0.10	68.12±0.22	81.53±0.19	54.21±0.16	57.02±0.35
Segmenter [56]	ViT-Tiny	91.42±0.16	95.21±0.13	85.34±0.22	85.12±0.19	88.16±0.14	89.05±0.11	66.25±0.24	80.52±0.31	51.54±0.21	56.32±0.45
BANet [50]	ResT-Lite	93.12±0.18	96.35±0.12	87.10±0.25	88.42±0.15	95.16±0.09	92.03±0.11	68.05±0.20	81.62±0.22	54.33±0.14	57.18±0.38
SwinUperNet [28]	Swin-Tiny	94.24±0.22	96.99±0.03	87.66±0.33	89.20±0.13	95.77±0.11	92.77±0.11	70.74±0.11	83.66±0.10	57.66±0.14	61.72±0.47
DC-Swin [49]	Swin-Small	94.49±0.10	97.12±0.06	88.19±0.08	89.37±0.17	96.04±0.15	93.04±0.04	71.17±0.07	83.80±0.07	58.46±0.07	61.16±0.31
UNetFormer [51]	ResNet18	94.36±0.07	96.78±0.07	87.56±0.02	88.64±0.16	96.23±0.08	92.71±0.02	70.79±0.14	82.95±0.31	58.11±0.22	58.15±0.62
AerialFormer-B [13]	Swin-Base	93.74±0.28	96.60±0.35	86.05±0.61	88.43±0.27	94.73±0.42	91.91±0.22	69.33±0.41	82.76±0.46	56.22±0.42	57.13±0.23
SFA-Net [16]	EfficientNet-B3	94.69±0.15	97.27±0.07	88.01±0.09	89.35±0.01	96.49±0.08	93.16±0.03	72.06±0.14	84.00±0.09	59.67±0.16	62.43±0.77
PyramidMamba [48]	Swin-Base	94.82±0.11	97.28±0.05	88.64±0.12	89.96±0.08	97.22±0.09	93.58±0.04	72.84±0.13	84.22±0.15	60.12±0.14	62.88±0.52
AFENet [10]	ResNet18	94.60±0.07	97.06±0.08	87.87±0.05	88.88±0.01	96.37±0.10	92.96±0.09	72.07±0.11	83.98±0.06	59.76±0.12	60.52±0.27
IGECNet [59]	Res2Net50	93.72±0.08	96.20±0.06	86.88±0.23	88.33±0.09	96.02±0.26	92.23±0.07	70.09±0.04	82.82±0.08	57.18±0.06	60.26±0.48
CFormer [53]	ResNet50	93.87±0.36	96.66±0.16	86.73±0.13	87.33±0.35	95.70±0.22	92.06±0.13	**73.99±0.16**	79.86±0.31	**62.11±0.24**	**67.41±1.44**
H-CoReNet (Ours)	EfficientNet-B3	95.63±0.17	97.94±0.13	89.91±0.24	90.13±0.21	96.82±0.16	94.09±0.09	73.41±0.18	84.47±0.23	61.36±0.21	59.43±0.63
H-CoReNet (Ours)	ConvNeXt-Base	**96.03±0.14**	**97.96±0.11**	**90.22±0.19**	**90.81±0.18**	**97.35±0.13**	**94.47±0.11**	73.81±0.16	**85.63±0.19**	61.89±0.17	66.17±0.56

TABLE IV

CLASSWISE BOUNDARY-SENSITIVE COMPARISON ON POTSDAM UNDER THE TTA PROTOCOL. B-F1, TRI-IOU, B-IOU, AND S-IOU ARE COMPUTED WITH THE SAME CLASS MASKS AND BOUNDARY TOLERANCE

Method	Imp.surf.				Building				Car			
	B-F1	Tri-IoU	B-IoU	S-IoU	B-F1	Tri-IoU	B-IoU	S-IoU	B-F1	Tri-IoU	B-IoU	S-IoU
DANet [9]	68.32±0.18	83.45±0.55	53.62±0.25	48.10±0.58	66.12±0.42	88.95±0.48	51.85±0.52	57.25±1.85	90.65±0.10	92.95±0.58	80.12±0.45	81.24±0.70
ABCNet [21]	70.12±0.24	83.95±0.65	55.98±0.21	48.75±0.95	67.23±0.42	89.12±0.48	54.12±0.38	56.11±1.45	91.85±0.18	94.12±0.45	83.65±0.29	85.12±0.85
Segmenter [56]	68.14±0.28	83.12±0.72	53.22±0.34	47.65±1.12	65.80±0.52	88.45±0.61	51.24±0.46	55.70±1.98	90.34±0.22	92.56±0.64	79.54±0.51	80.54±0.95
BANet [50]	70.21±0.20	84.10±0.58	56.12±0.18	48.90±0.85	67.45±0.35	89.32±0.42	54.38±0.31	56.40±1.35	92.10±0.14	94.28±0.38	83.85±0.25	85.35±0.72
SwinUperNet [28]	73.51±0.23	85.30±0.38	59.80±0.15	52.50±1.41	72.41±0.21	92.02±0.53	60.80±0.19	65.53±1.83	92.93±0.06	95.44±0.21	85.73±0.57	88.00±0.70
DC-Swin [49]	74.15±0.08	84.72±0.42	60.34±0.28	50.95±1.05	73.67±0.30	91.29±0.22	61.64±0.28	62.95±3.03	92.59±0.27	96.40±0.31	87.55±0.19	89.16±1.10
UNetFormer [51]	73.49±0.23	84.45±0.64	60.30±0.43	49.76±1.05	72.53±0.27	90.87±0.62	60.69±0.44	56.84±1.54	93.75±0.11	95.58±0.51	88.11±0.24	86.59±1.68
AerialFormer-B [13]	72.42±0.48	84.83±0.81	58.84±0.85	50.59±2.27	72.13±0.54	88.75±1.60	58.74±0.69	53.02±8.15	91.20±0.94	96.17±0.79	84.28±0.57	88.00±2.80
SFA-Net [16]	74.56±0.25	85.91±0.58	61.72±0.20	54.45±1.48	74.50±0.14	91.00±0.11	63.20±0.29	63.13±2.37	94.11±0.12	96.57±0.27	89.26±0.07	90.06±0.95
PyramidMamba [48]	74.12±0.18	85.46±0.42	61.10±0.16	53.80±1.22	74.15±0.20	91.12±0.35	62.84±0.22	62.90±2.14	93.84±0.15	96.12±0.32	88.64±0.12	89.46±0.85
AFENet [10]	75.16±0.13	85.69±0.48	62.22±0.30	51.03±0.98	74.35±0.23	91.72±0.48	63.20±0.29	60.48±1.37	94.01±0.09	96.86±0.13	89.33±0.15	90.14±0.62
IGECNet [59]	72.46±0.06	84.40±0.35	58.73±0.10	51.07±0.94	71.35±0.05	89.57±0.56	58.74±0.22	57.76±1.36	93.65±0.23	96.26±0.22	88.35±0.31	90.03±0.57
CFormer [53]	76.39±0.30	83.35±1.04	**64.93±0.47**	**59.07±3.64**	**76.37±0.53**	84.93±0.98	**66.24±0.56**	64.67±4.04	94.15±0.29	95.11±0.70	89.88±0.42	**96.45±0.72**
H-CoReNet (Ours)	76.24±0.29	85.91±0.63	63.78±0.41	48.53±1.31	76.17±0.37	91.66±0.59	65.82±0.47	57.21±1.97	**94.91±0.21**	95.78±0.49	89.96±0.34	88.23±1.08
H-CoReNet (Ours)	**76.83±0.25**	**87.08±0.57**	64.78±0.35	52.14±1.19	75.69±0.33	**92.51±0.52**	65.73±0.42	**66.08±1.76**	94.38±0.19	**97.29±0.41**	**90.61±0.29**	92.06±0.91

TABLE V

QUANTITATIVE COMPARISON ON THE VAIHINGEN TEST SET UNDER THE TTA PROTOCOL

Method	Backbone	Imp.surf.	Building	LowVeg	Tree	Car	mF1↑	mB-F1	mTri-IoU	mB-IoU	mS-IoU
DANet [9]	ResNet18	90.35±0.35	94.22±0.28	82.64±0.42	87.71±0.31	46.93±1.15	80.37±0.25	41.52±0.58	70.35±0.62	30.54±0.48	50.12±1.22
ABCNet [21]	ResNet18	95.24±0.12	94.15±0.08	83.12±0.14	88.35±0.10	84.77±0.25	89.13±0.09	70.62±0.15	81.24±0.22	55.35±0.18	65.72±0.45
Segmenter [56]	ViT-Tiny	92.42±0.18	95.35±0.14	84.06±0.25	90.12±0.16	87.14±0.32	89.82±0.12	68.24±0.25	81.35±0.34	54.12±0.28	57.65±0.85
BANet [50]	ResT-Lite	94.79±0.07	89.75±0.13	80.09±0.38	87.86±0.10	68.83±0.42	84.26±0.18	58.43±0.17	73.33±0.42	41.84±0.19	60.43±0.67
SwinUperNet [28]	Swin-Tiny	96.87±0.16	95.66±0.33	84.26±0.22	89.86±0.18	87.80±0.62	90.89±0.16	72.23±0.26	82.03±1.03	57.08±0.95	67.86±1.57
DC-Swin [49]	Swin-Small	97.03±0.02	96.17±0.07	85.08±0.20	90.28±0.08	89.17±0.24	91.55±0.06	72.80±0.10	83.24±0.12	59.02±0.05	66.89±0.82
UNetFormer [51]	ResNet18	96.91±0.04	95.80±0.10	84.86±0.15	90.17±0.04	89.10±0.28	91.37±0.06	72.48±0.15	82.73±0.03	58.57±0.14	66.19±0.35
SFA-Net [16]	EfficientNet-B3	95.84±0.07	95.94±0.12	84.19±0.08	89.74±0.10	89.37±0.47	91.02±0.10	71.84±0.27	82.61±0.12	57.79±0.38	67.37±0.61
PyramidMamba [48]	Swin-Base	96.88±0.08	95.92±0.07	85.31±0.14	90.11±0.09	88.93±0.22	91.43±0.05	71.80±0.15	82.95±0.11	58.02±0.16	68.12±0.55
D2LS [70]	ConvNeXt-Base	97.01±0.03	95.86±0.06	84.94±0.10	90.32±0.05	90.24±0.17	91.67±0.04	71.82±0.13	83.29±0.13	58.26±0.10	68.61±0.76
AFENet [10]	ResNet18	96.94±0.07	96.06±0.16	84.94±0.30	90.33±0.08	89.88±0.29	91.63±0.11	73.96±0.14	83.34±0.18	60.53±0.14	66.81±0.95
IGECNet [59]	Res2Net50	96.64±0.04	95.32±0.11	84.49±0.13	90.20±0.09	89.33±0.21	91.20±0.07	72.34±0.18	82.30±0.16	58.21±0.30	65.83±0.41
CFormer [53]	ResNet50	96.62±0.10	95.66±0.13	84.99±0.11	89.94±0.12	84.78±1.27	90.40±0.24	**79.60±0.35**	78.74±0.42	**63.23±0.47**	70.51±0.80
H-CoReNet (Ours)	EfficientNet-B3	**97.14±0.08**	96.43±0.11	84.91±0.17	90.21±0.12	**91.47±0.36**	92.03±0.09	74.11±0.16	83.58±0.21	60.69±0.24	66.17±0.73
H-CoReNet (Ours)	ConvNeXt-Base	97.06±0.06	**96.44±0.09**	**85.32±0.14**	**90.43±0.08**	91.02±0.28	**92.05±0.13**	74.13±0.14	**84.39±0.18**	61.08±0.19	**71.06±0.67**

pedestrians [42]. D2LS recovers only parts of these instances, whereas SFA-Net and UNetFormer show structural omissions. The traffic sign in the upperleft corner is also challenging: UNetFormer labels it as a building, while SFA-Net and D2LS omit it. H-CoReNet preserves a more complete pedestrian contour and retains the traffic sign in this example. These observations concern the complete model; the contribution of each component is assessed separately by the controlled ablations and module visualizations.

Fig. 5(b) displays densely distributed static vehicles. The comparison models frequently miss individual vehicles or confuse trucks with buildings, whereas H-CoReNet produces more complete and better separated vehicle masks in the displayed scene. Because this figure compares final predictions rather than intermediate counterfactual features, the scene-level improvement is treated as whole-model evidence rather than attributed to a single module.

2) Potsdam Dataset Experimental Results: Experiments evaluate urban landscape parsing on the ISPRS Potsdam dataset. This dataset possesses a 5-cm spatial resolution. Such high resolution introduces complex geometric detail noise. Object boundaries of artificial facilities, such as buildings and impervious surfaces, often blur. Natural landscapes, including trees and low vegetation, overlap in the spatial frequency domain. Densely distributed tiny vehicles face interference from complex background gradients [35]. Large-capacity

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TABLE VI

CLASSWISE BOUNDARY-SENSITIVE COMPARISON ON VAIHINGEN UNDER THE TTA PROTOCOL. B-F1, TRI-IOU, B-IOU, AND S-IOU ARE COMPUTED WITH THE SAME CLASS MASKS AND BOUNDARY TOLERANCE

Method	Imp.surf.				Building				Car			
	B-F1	Tri-IoU	B-IoU	S-IoU	B-F1	Tri-IoU	B-IoU	S-IoU	B-F1	Tri-IoU	B-IoU	S-IoU
DANet [9]	42.35±0.72	70.12±1.15	28.54±0.65	41.22±1.84	33.46±0.85	72.34±1.42	21.22±0.58	38.45±2.14	61.30±1.24	80.54±1.68	42.35±0.92	71.04±1.95
ABCNet [21]	72.93±0.46	85.96±0.40	60.38±0.45	46.92±3.03	76.25±0.28	87.41±0.68	62.67±0.62	64.73±3.65	83.78±0.20	83.24±0.94	67.82±0.72	86.83±0.81
Segmenter [56]	71.52±0.47	84.74±0.61	59.02±0.39	45.19±2.27	74.58±0.36	87.48±0.57	60.99±0.43	62.17±2.83	85.71±0.73	86.26±0.95	71.61±0.67	88.13±1.33
BANet [50]	62.71±0.27	81.16±0.16	48.01±0.31	50.31±0.81	60.01±0.21	78.61±0.42	44.43±0.30	55.30±1.76	61.03±0.31	64.87±1.60	42.71±0.35	68.48±1.74
SwinUperNet [28]	74.14±1.16	85.56±1.03	61.78±1.02	50.62±3.95	78.58±0.44	88.59±2.39	64.86±1.98	66.31±6.70	85.65±0.46	85.74±2.93	69.97±2.44	88.66±2.62
DC-Swin [49]	75.30±0.15	85.93±0.60	63.14±0.19	46.90±1.15	79.76±0.36	89.39±0.33	67.33±0.21	64.42±1.83	84.68±0.49	89.62±0.43	74.17±0.30	92.51±0.29
UNetFormer [51]	74.42±0.10	85.66±0.54	62.43±0.10	44.76±1.51	78.58±0.44	89.73±0.53	66.41±0.23	64.38±1.88	86.27±0.41	87.88±0.10	74.10±0.53	90.88±0.33
SFA-Net [16]	74.80±0.32	86.54±0.22	63.18±0.47	52.40±1.66	79.68±0.34	90.06±0.39	67.53±0.52	66.83±3.10	88.53±0.64	90.62±0.49	77.69±0.40	92.94±0.29
PyramidMamba [48]	74.22±0.25	86.10±0.18	62.45±0.20	50.85±1.45	78.15±0.22	89.35±0.40	65.80±0.33	67.90±2.50	87.20±0.45	90.85±0.42	76.50±0.38	93.25±0.35
D2LS [70]	73.95±0.27	85.85±0.21	62.05±0.11	49.68±1.51	77.19±0.29	88.99±0.53	64.75±0.27	68.84±2.00	86.17±0.38	91.08±0.55	75.81±0.47	93.75±0.49
AFENet [10]	75.80±0.12	**86.56±0.54**	64.45±0.17	44.96±4.13	80.49±0.48	90.07±0.31	69.03±0.25	65.80±0.56	88.35±0.29	88.65±0.65	76.91±0.44	91.78±0.46
IGECNet [59]	73.71±0.30	85.50±0.47	61.72±0.38	46.97±3.56	78.01±0.43	88.06±0.74	64.84±0.54	62.27±3.35	87.39±0.54	87.14±0.71	75.15±0.78	90.13±0.60
CFormer [53]	**80.62±0.27**	84.65±0.58	**67.46±0.58**	**59.40±4.68**	**83.75±0.92**	86.31±0.51	**71.28±0.44**	**72.75±2.06**	87.26±1.00	77.54±3.15	73.33±2.12	79.18±3.73
H-CoReNet (Ours)	76.11±0.31	85.58±0.64	64.37±0.42	42.46±1.81	80.76±0.43	90.27±0.52	69.24±0.48	64.07±2.13	**89.54±0.36**	90.14±0.61	78.23±0.51	91.96±0.94
H-CoReNet (Ours)	76.69±0.27	85.47±0.56	64.66±0.35	50.97±1.46	80.47±0.39	**91.44±0.47**	70.23±0.41	71.84±1.72	88.74±0.29	**92.36±0.49**	**78.86±0.44**	**94.17±0.71**

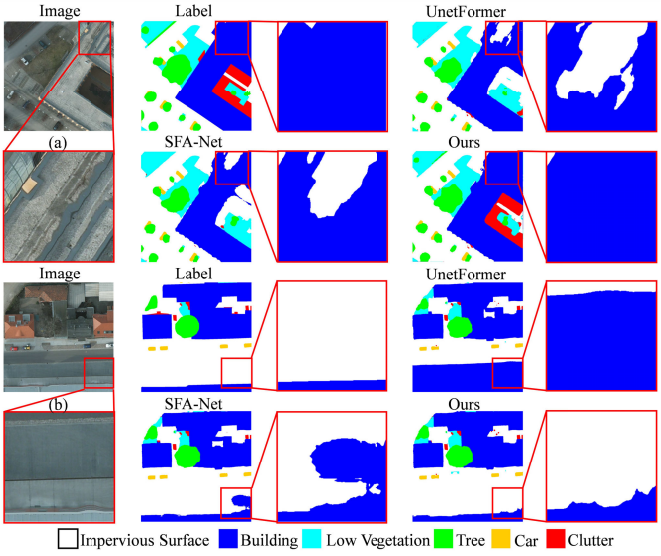


Fig. 6. Native-resolution qualitative comparison using selected regions from 6000 × 6000 Potsdam orthophotos. The magnified views preserve the surrounding scene context. (a) H-CoReNet restores building continuity and reduces prediction leakage into clutter. (b) Better separates impervious surfaces from buildings and retains more coherent object boundaries than the compared predictions.

networks exhibit macroscopic structural deformations in these scenarios.

Table III presents the repeated-run TTA comparison, and Table IV reports classwise boundary-sensitive metrics. H-CoReNet with ConvNeXt-Base obtains 94.47% ±0.11% mF1; the EfficientNet-B3 configuration obtains 94.09% ±0.09%. The ConvNeXt-Base configuration further records 73.81% ±0.16% mB-F1, 85.63% ±0.19% mTri-IoU, 61.89% ±0.17% mB-IoU, and 66.17% ±0.56% mS-IoU. These repeated-run estimates are used for the quantitative claims, while the qualitative cases in Fig. 6 illustrate the corresponding boundary and small-object behavior.

Fig. 6 presents qualitative comparisons on scenes containing densely distributed impervious surfaces and buildings with similar local appearance. In Fig. 6(a), several comparison methods produce structural breaks in the building masks, whereas H-CoReNet retains more continuous building regions and shows less leakage into adjacent impervious surfaces and clutter. This visual comparison records the final prediction behavior of the complete network.

Fig. 6(b) shows regions with strong interclass appearance overlap. UNetFormer labels a large road region as building, and SFA-Net retains smaller mixed predictions. In the enlarged region on the right, several baselines also confuse small buildings with impervious surfaces. H-CoReNet produces more coherent interiors and preserves the displayed building contours more consistently. Module-specific effects are evaluated by the repeated-run ablation, rather than inferred from this single scene.

3) Vaihingen Dataset Experimental Results: The Vaihingen dataset serves for spatial parsing validation. This dataset possesses a 9-cm ground sample distance (GSD). Original images fuse near-infrared, red, and green bands. Under this spectrum, buildings and impervious surfaces exhibit high representation similarities [6]. Background features often submerge sparse tiny vehicles. Conventional networks often struggle to separate overlapping feature responses when processing nonvisible bands. Semantic aliasing occurs among densely intertwined urban entities.

Table V reports the repeated-run TTA comparison, and Table VI reports classwise boundary-sensitive metrics. H-CoReNet obtains 92.03% ± 0.09% mF1 with EfficientNet-B3 and 92.05% ± 0.13% with ConvNeXt-Base, compared with 91.67% ± 0.04% for D2LS. The ConvNeXt-Base setting records 74.13% ± 0.14% mB-F1, 84.39% ± 0.18% mTri-IoU, 61.08% ± 0.19% mB-IoU, and 71.06% ± 0.67% mS-IoU. The EfficientNet-B3 configuration attains 91.47% ± 0.36% $F1$ for cars and 97.14% ± 0.08% for impervious surfaces.

The qualitative results in Fig. 7 reflect the difficulty of mapping small objects and adjacent classes in IRRG imagery. In Fig. 7(a), SFA-Net and D2LS produce structural breaks in building predictions, whereas H-CoReNet retains more continuous building regions. Fig. 7(b) contains complex building edges and small vehicles obscured by illumination and haze. The comparison masks show boundary leakage or recover only partial vehicle contours, while H-CoReNet produces a smoother building boundary and more localized vehicle regions. These are whole-model observations; the roles of

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TABLE VII

QUANTITATIVE COMPARISON ON THE LOVEDA TEST SET UNDER THE TTA PROTOCOL

Method	Backbone	Background	Building	Road	Water	Barren	Forest	Agriculture	mIoU↑	mB-F1	mTri-IoU	mB-IoU	mS-IoU
TransUNet [1]	ResNet50	42.82±0.53	54.79±0.92	52.54±1.32	78.18±0.54	8.86±1.42	45.39±0.71	54.86±1.76	48.21±0.68	14.18±0.48	29.05±0.74	4.43±0.17	20.96±0.58
DC-Swin [49]	Swin-Tiny	43.70±1.14	58.47±1.12	57.29±1.01	80.20±0.77	19.07±1.35	47.85±0.60	58.99±1.95	52.22±0.60	20.05±0.12	33.73±0.63	6.39±0.04	22.72±0.42
UNetFormer [51]	ResNet18	43.45±0.47	55.14±1.91	54.37±0.98	77.83±0.44	17.28±1.23	46.47±0.88	57.92±1.97	50.35±0.85	15.94±0.57	30.66±0.73	4.99±0.21	22.16±0.61
Hi-ResNet	Hi-ResNet	46.76±0.58	57.62±1.04	55.00±0.93	79.82±0.52	17.19±1.55	47.10±0.71	60.87±1.48	52.05±0.45	26.84±0.24	40.68±0.44	8.75±0.08	25.91±0.41
AerialFormer [13]	Swin-Base	46.84±0.42	59.70±1.12	58.34±0.94	81.01±0.40	18.29±1.08	48.42±0.44	61.34±1.19	53.42±0.47	29.46±0.18	43.08±0.31	9.93±0.06	27.42±0.34
SFA-Net [16]	EfficientNet-B3	46.03±0.58	60.05±0.81	56.25±1.83	81.38±0.56	21.08±2.66	47.09±0.73	64.09±1.16	53.71±0.49	30.59±0.20	44.14±0.29	10.33±0.07	26.95±0.29
D2LS [70]	ConvNeXt-Base	47.42±0.29	61.30±0.25	58.55±0.57	81.47±0.17	**23.25±1.52**	48.95±0.30	64.02±0.55	54.99±0.27	30.02±0.14	43.46±0.23	10.21±0.05	27.76±0.37
AFENet [10]	ResNet18	46.74±0.44	60.81±0.67	58.57±2.08	82.38±0.30	20.08±1.14	49.24±0.22	63.96±0.70	54.54±0.26	**34.91±0.14**	**46.65±0.17**	**12.03±0.06**	29.15±0.47
CFormer [53]	ResNet50	**48.67±0.39**	58.55±0.99	58.50±1.23	81.80±0.27	16.21±1.67	44.64±2.84	**67.85±0.51**	53.75±0.61	28.83±0.57	41.59±0.41	9.65±0.21	**33.26±1.18**
H-CoReNet (Ours)	EfficientNet-B3	47.13±0.42	59.62±0.73	59.18±1.16	81.53±0.37	17.21±1.28	**49.91±0.46**	65.71±0.82	54.33±0.21	27.58±0.19	42.94±0.27	9.11±0.13	27.14±0.61
H-CoReNet (Ours)	ConvNeXt-Base	48.13±0.31	**62.71±0.39**	**61.18±0.62**	**82.73±0.21**	20.58±1.13	49.71±0.34	66.51±0.48	**55.94±0.16**	32.27±0.16	45.47±0.22	10.93±0.11	29.38±0.53

TABLE VIII

CLASSWISE BOUNDARY-SENSITIVE COMPARISON ON LOVEDA UNDER THE TTA PROTOCOL. B-F1, TRI-IOU, B-IOU, AND S-IOU ARE COMPUTED WITH THE SAME CLASS MASKS AND BOUNDARY TOLERANCE

Method	Building				Road			
	B-F1	Tri-IoU	B-IoU	S-IoU	B-F1	Tri-IoU	B-IoU	S-IoU
TransUNet [1]	20.50±0.88	43.00±1.55	6.15±0.28	51.80±3.92	20.80±1.20	22.00±1.68	6.70±0.42	3.80±0.69
DC-Swin [49]	29.15±0.52	46.64±1.64	9.27±0.18	48.89±4.45	28.16±0.84	30.67±1.59	9.51±0.33	4.69±0.84
UNetFormer [51]	22.19±1.06	44.88±1.89	6.84±0.34	54.43±4.13	23.09±1.60	23.87±2.04	7.65±0.58	4.35±0.53
Hi-Resnet [4]	34.60±0.31	48.20±0.83	11.20±0.12	57.10±2.10	41.70±0.44	43.90±0.78	15.10±0.18	13.10±1.08
AerialFormer [13]	37.40±0.24	49.60±0.62	12.40±0.09	58.90±1.45	46.80±0.26	48.60±0.55	17.50±0.10	16.60±0.67
SFA-Net [16]	37.69±0.19	49.08±0.69	12.44±0.09	55.27±1.14	49.31±0.29	50.81±0.55	18.81±0.17	17.44±1.47
D2LS [70]	38.76±0.20	50.61±0.59	12.95±0.08	59.75±1.62	46.42±0.15	48.39±0.48	17.56±0.05	16.49±0.40
AFENet [10]	**43.05±0.26**	**52.32±0.41**	**14.63±0.12**	62.50±0.98	**53.11±0.26**	**54.37±0.37**	**20.70±0.19**	18.78±1.23
CFormer [53]	36.97±0.96	51.41±0.81	12.13±0.38	**67.26±1.22**	44.81±0.64	43.77±0.47	17.02±0.31	13.83±2.17
H-CoReNet (Ours)	33.76±0.48	48.58±0.71	10.91±0.26	56.21±1.84	43.16±0.62	47.53±0.83	15.98±0.41	16.19±1.12
H-CoReNet (Ours)	41.52±0.35	52.03±0.58	13.97±0.19	59.91±1.47	49.17±0.44	51.28±0.64	18.68±0.31	**19.37±0.96**

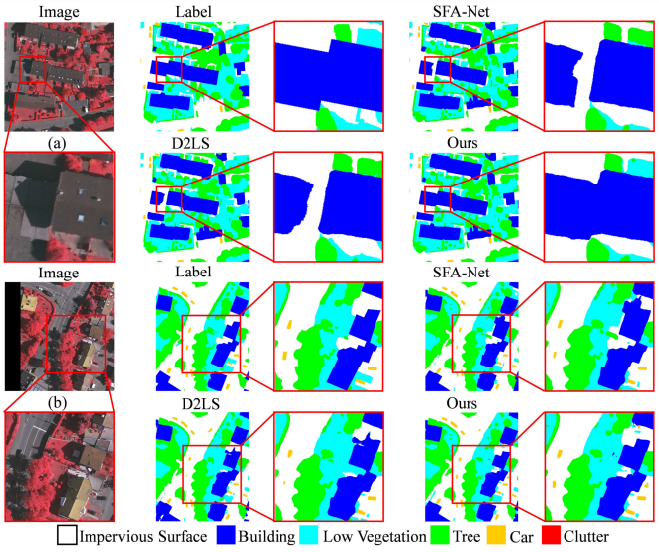


Fig. 7. Native-resolution qualitative comparison using selected regions from large Vaihingen orthophotos. The magnified views retain the original near-infrared urban context. (a) H-CoReNet preserves the continuity of dense urban structures. (b) It retains sparse vehicle regions and their contours more consistently under the displayed illumination and haze conditions.

LDRWA, HCU, and SFO are supported separately by ablation and dedicated feature visualizations. Additional large-scale qualitative results are provided for Potsdam and Vaihingen (Fig. 8).

4) LoveDA Dataset Experimental Results: The LoveDA dataset contains urban and rural landscapes. Scene switching triggers cross-domain distribution shifts. Class imbalance constitutes a significant parsing challenge. Test set ground truth labels remain unreleased. Inference results are submitted to the official server for online evaluation.

Table VII reports the repeated-run TTA comparison, and Table VIII reports classwise boundary-sensitive metrics. H-CoReNet with ConvNeXt-Base obtains 55.94% ± 0.16% mIoU, compared with 54.99% ± 0.27% for D2LS; EfficientNet-B3 obtains 54.33% ± 0.21%. The ConvNeXt-Base setting records 32.27% ± 0.16% mB-F1, 45.47% ± 0.22% mTri-IoU, 10.93% ± 0.11% mB-IoU, and 29.38% ± 0.53% mS-IoU. The 66.51% ± 0.48% agriculture IoU indicates that cross-domain texture variation remains challenging despite the overall gain.

Fig. 9 reflects the remaining difficulty of cross-domain parsing. At the complex rural boundaries in Fig. 9(a), UNetFormer and SFA-Net confuse large background regions with agricultural areas, while D2LS produces fragmented boundary predictions. H-CoReNet reduces this leakage and preserves a more continuous boundary in the displayed scene. The figure is interpreted as whole-model evidence; the operation of SFO is examined independently in Supplementary Fig. 6.

Fig. 9(b) presents a cross-domain case in which dense intersections and building roofs have similar reflectance. UNetFormer loses parts of the urban topology, SFA-Net assigns portions of building clusters to background, and D2LS misses elongated road segments and merges adjacent building boundaries. H-CoReNet preserves more of the road network and separates the displayed building contours more consistently. This qualitative result is interpreted as whole-model evidence and is not assigned to an unobserved frequency decomposition within HCU.

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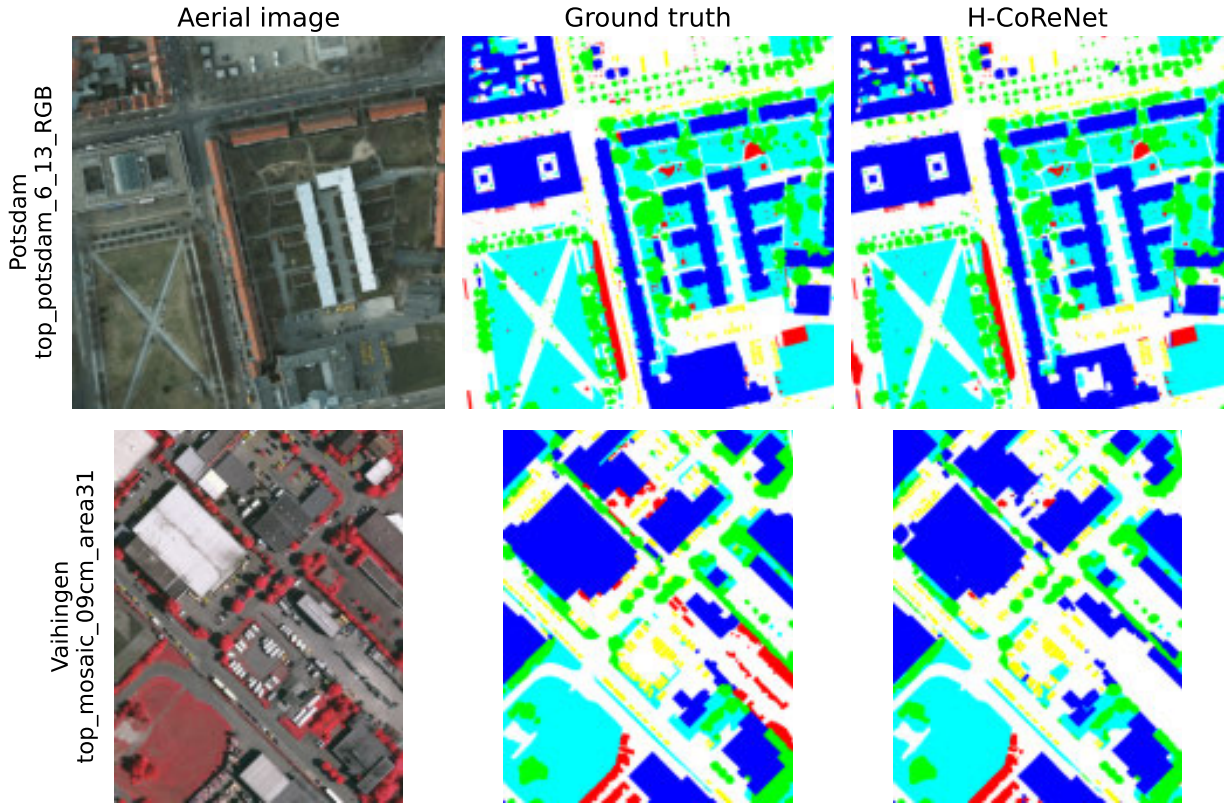


Fig. 8. Large-scale qualitative evaluation on selected native-resolution test regions from Potsdam and Vaihingen. Each row presents the aerial image, ground-truth mask, and H-CoReNet prediction. The Potsdam example is drawn from a 6000 × 6000 tile, and the Vaihingen example is drawn from a large IRRG orthophoto. The panels retain broad scene context while exposing building boundaries, impervious surfaces, vegetation, and small vehicles.

TABLE IX

UNIFIED CORE-MODULE ABLATION AND PARAMETER ANALYSIS. REPEATED-RUN SEGMENTATION ACCURACY AND PARAMETER COUNTS ARE ALIGNED BY MODULE COMBINATION. SFO, HCU, AND LDRWA DENOTE THE SFO, HCU, AND LDRWA, RESPECTIVELY

LDRWA	HCU	SFO	Potsdam (mF1 %)	Vaihingen (mF1 %)	LoveDA (mIoU %)	UAVid (mIoU %)	ConvNeXt-Base Params (M)	EfficientNet-B3 Params (M)
×	×	×	93.27±0.13	91.79±0.09	55.26±0.16	69.62±0.18	89.30	11.70
✓	×	×	93.75±0.11	91.92±0.08	55.60±0.13	70.06±0.15	89.40	11.80
×	✓	×	93.68±0.12	91.88±0.10	55.52±0.14	70.24±0.17	89.40	11.80
×	×	✓	93.61±0.14	91.85±0.09	55.46±0.15	70.69±0.14	89.40	11.80
✓	✓	×	94.07±0.10	91.98±0.07	55.75±0.12	70.83±0.13	89.50	11.90
✓	×	✓	94.14±0.09	91.96±0.08	55.81±0.11	71.15±0.12	89.50	11.90
×	✓	✓	94.01±0.12	91.93±0.10	55.71±0.13	71.27±0.14	89.60	12.00
✓	✓	✓	**94.47±0.11**	**92.05±0.13**	**55.94±0.16**	**71.54±0.17**	89.70	12.10

E. Ablation Study

1) Ablation on Core Mechanisms: Ablation experiments evaluate core modules on four benchmark datasets. These tests verify the spatial parsing capabilities of each component. The baseline model retains only backbone feature extraction and basic spatial upsampling paths. This configuration ensures evaluation objectivity. Table IX records mapping accuracy evolution after introducing core components.

Table IX reports mean and standard deviation under the same TTA protocol. Relative to the backbone-only configuration, the complete model increases Potsdam mF1 from 93.27% ± 0.13% to 94.47% ± 0.11%, Vaihingen mF1 from 91.79% ± 0.09% to 92.05% ± 0.13%, LoveDA mIoU from 55.26% ± 0.16% to 55.94% ± 0.16%, and UAVid mIoU from 69.62% ± 0.18% to 71.54% ± 0.17%.

The single-module rows separate the contributions. On UAVid, LDRWA, HCU, and SFO reach 70.06% ± 0.15%, 70.24% ± 0.17%, and 70.69% ± 0.14% mIoU, respectively. The LDRWA+SFO combination reaches 71.15% ± 0.12%, and the complete model reaches 71.54% ± 0.17%. The smaller gain on Vaihingen is reported with its run-to-run variation rather than interpreted from a single seed.

Table IX reports accuracy and parameter growth in a unified rowwise comparison. With ConvNeXt-Base, the complete configuration contains 89.70M parameters, compared with 89.30M for the backbone-only configuration; with EfficientNet-B3, the corresponding values are 12.10 and 11.70M. Thus, the three modules add 0.40M parameters under either backbone. Complete-network FLOPs and latency are reported separately under the matched UAVid protocol in Table X.

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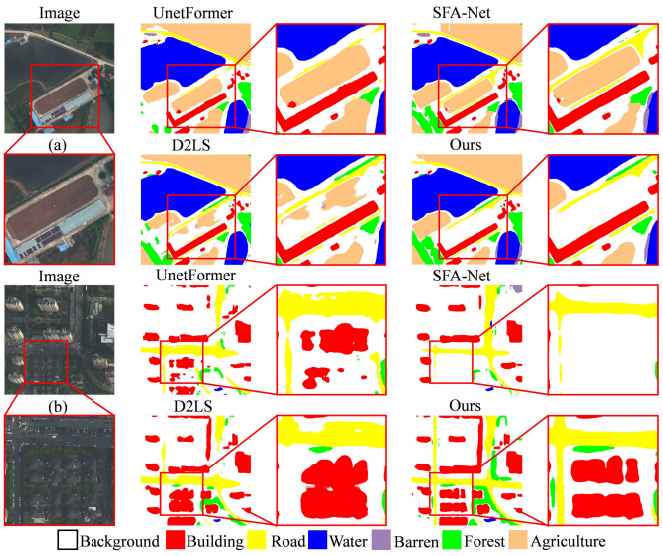


Fig. 9. Visual comparison of semantic segmentation results on the LoveDA test set. (a) H-CoReNet reduces confusion between background and agricultural regions in the displayed rural scene. (b) It preserves more continuous roads and better separates adjacent building roofs in the displayed urban scene.

TABLE X

MATCHED COMPUTATIONAL-EFFICIENCY COMPARISON ON UAVID. FLOPS AND LATENCY ARE MEASURED USING ONE 512 × 512 INPUT ON AN RTX 4090 WITH FP32 INFERENCE

Method	Params (M)	FLOPs (G)	FP32 latency (ms)
AerialFormer-B [13]	113.8	126.8	19.9 ± 0.7
SFA-Net [16]	10.7	7.1	15.0 ± 3.0
D2LS [70]	90.9	89.2	10.0 ± 1.4
H-CoReNet (Ours)	89.7	100.6	12.4 ± 0.5

Table X reports complete-network parameter count, FLOPs, and FP32 latency for one 512 × 512 UAVid input. Following the convention used in the original papers and official implementations, the operation count returned by THOP is reported directly: one multiply–accumulate is counted as one FLOP-equivalent operation, and no factor-of-two conversion is applied. Under this convention, H-CoReNet requires 89.7M parameters, 100.6G FLOPs, and 12.4 ± 0.5 ms per image. The comparison-method FLOP values are retained from their papers or official implementations under the same nominal input size; they are not described as outputs of one common profiler because custom attention operators can be counted differently. D2LS is reported at 90.9M parameters, 89.2G FLOPs, and 10.0 ± 1.4 ms. The table therefore supports an accuracy–cost tradeoff rather than a universal latency advantage.

2) Hyperparameter Sensitivity Analysis: Experiments first evaluate the LDRWA window size on UAVid. The 7 × 7 window provides the best measured mIoU in the supplied ablation table, balancing spatial coverage and background contamination. The Supplementary Material section "Hyperparameter Sensitivity Analysis" reports the classwise window-size results.

3) Dictionary-Size Definition and Sensitivity: The HCU dictionary capacity is determined by (10). To evaluate its sensitivity directly, we trained 30 independent models on Potsdam using $K \in \{12, 24, 32, 48, 64, 96\}$ and five random seeds $\{0, 42, 84, 126, 168\}$, and evaluated each checkpoint both with and without TTA. Fig. 10 shows a broad performance plateau across the tested dictionary sizes. Without TTA, the mean mF1 ranges from 94.05% to 94.12%, corresponding to a range of only 0.07 percentage points, while the mean mIoU ranges from 88.99% to 89.07%, a range of 0.08 percentage points. With TTA, the mean mF1 ranges from 94.41% to 94.47%, and the mean mIoU ranges from 89.70% to 89.80%, corresponding to ranges of 0.06 and 0.10 points, respectively. The default $K = 48$ achieves the highest mean mF1 and mIoU under both inference protocols: 94.12% ± 0.10% mF1 and 89.07% ± 0.17% mIoU without TTA, and 94.47% ± 0.11% mF1 and 89.80% ± 0.18% mIoU with TTA. The variations across dictionary sizes are comparable to the observed interseed variation, suggesting limited sensitivity to K over the tested interval. The default $K = 48$ lies on this broad plateau and avoids the larger dictionary of the $K = 64$ and $K = 96$ settings. We therefore retain $K = 48$ for Potsdam as a practical capacity–cost compromise, without interpreting the small mean differences as evidence of statistical superiority.

4) HCU Dictionary-Basis Interpretability: To examine how the shared HCU dictionary is used spatially, we generated basis-conditioned mode atlases for all 504 Potsdam and 113 Vaihingen test images using the seed-42 ConvNeXt-Base checkpoints. Each learned basis is a 128-D prototype, so the atlas visualizes its spatial response rather than imposing an artificial 2-D topology on the basis vector. Fig. 11 presents Potsdam test sample 0012 (tile `top_potsdam_4_13_0_14`). The top row contains the input image, ground-truth mask, TTA prediction, basis-relation graph, and role-coding legend. The lower panels display all 48 learned basis responses together with class-conditioned basis signatures. The responses are grouped post hoc by their observed dominant semantic role; this organization is used only for interpretation and does not impose a fixed one-to-one mapping between semantic classes and dictionary bases. The complete experiment produced 617 atlases.

5) Five-Seed Stability Evaluation: The stability protocol uses seeds $\{0, 42, 84, 126, 168\}$ with identical optimization and inference settings. For completed repeated runs, the tables report the mean and standard deviation of region-level and boundary-sensitive metrics. This protocol quantifies training variability independently of the environmental stress tests.

6) Robustness Evaluation Under Environmental Interference: The robustness analysis separates training-seed variability from environmental interference. In addition to the haze stress test, H-CoReNet and D2LS are evaluated from their seed-42 checkpoints on the Potsdam and Vaihingen test sets under the same TTA protocol. Five stronger perturbation families are applied without retraining: spectral crosstalk, spectral desaturation, exposure shift, gamma-response shift, and shadow attenuation. Averaged over these families, the normalized robustness area of H-CoReNet exceeds D2LS by 2.48 percentage points on Potsdam and 2.92 points on Vaihingen. The largest rAUC gaps occur for gamma shift on Potsdam (+7.86 points) and exposure shift on Vaihingen

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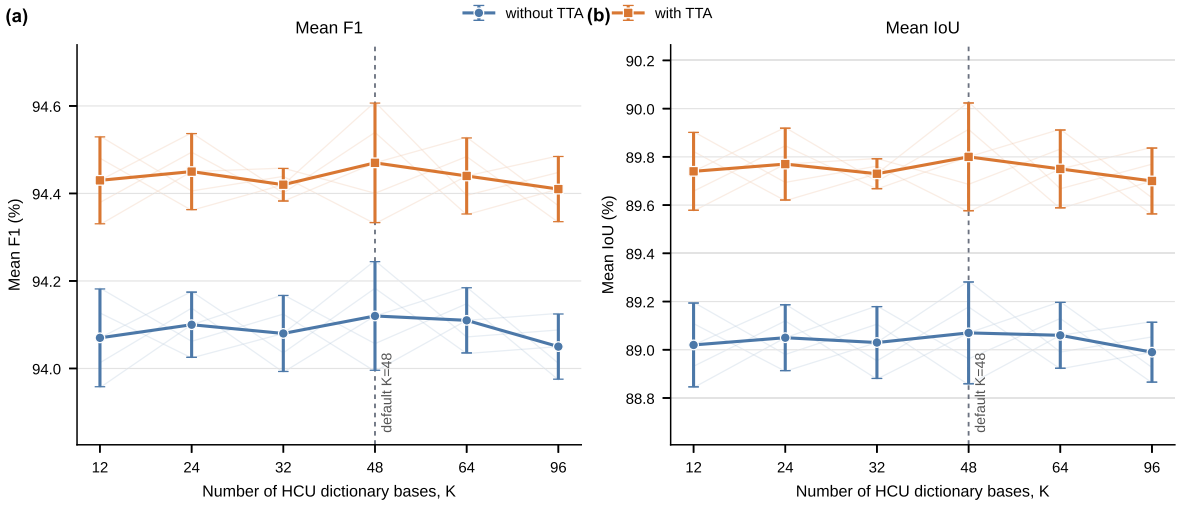


Fig. 10. Potsdam HCU dictionary-size sensitivity over five random seeds. (a) mF1. (b) mIoU. Lines show the mean, and error bars show one standard deviation for no-TTA and TTA evaluation. The dashed line marks the default $K = 48$.

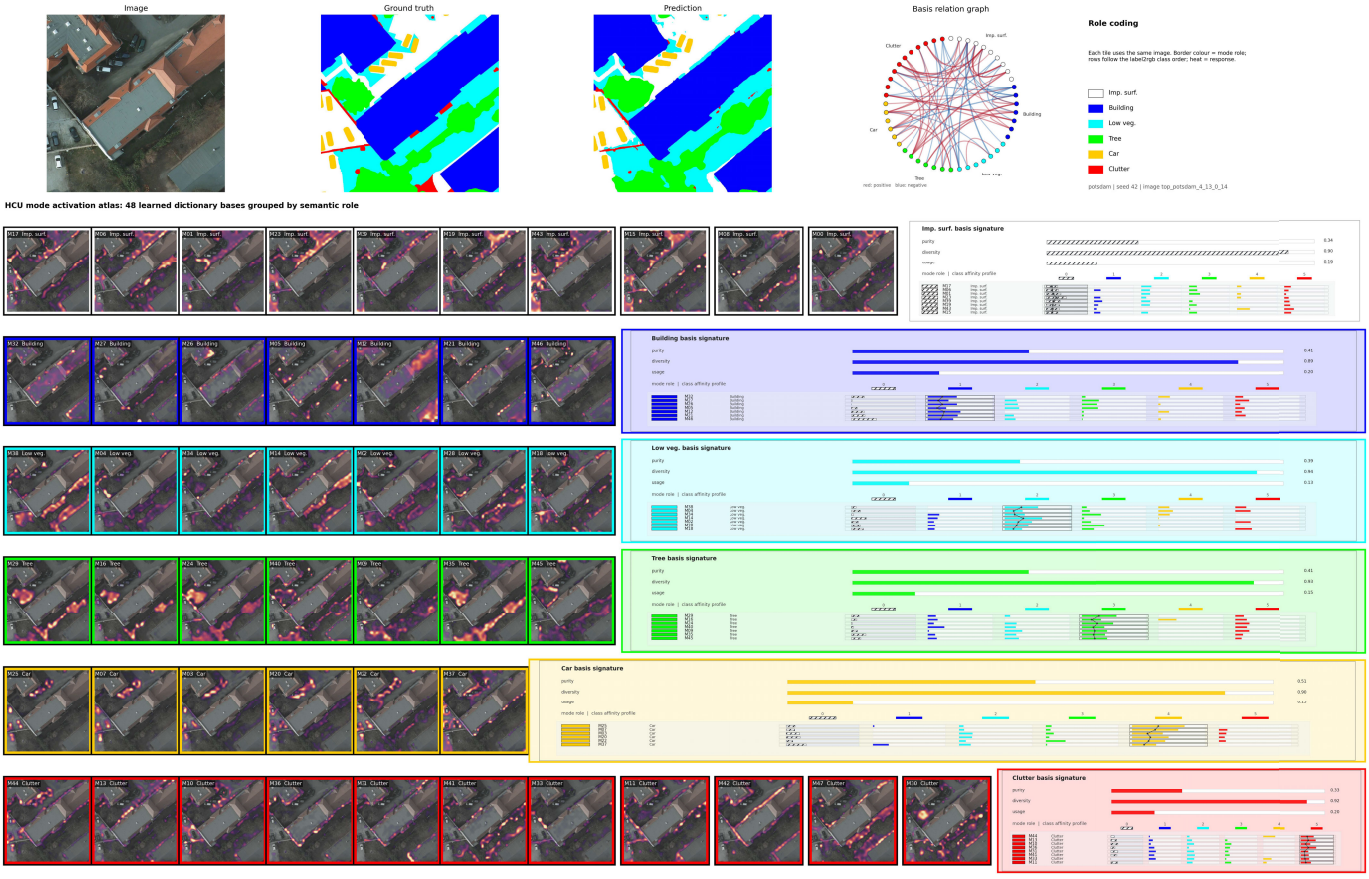


Fig. 11. Selected HCU dictionary-basis mode atlas for Potsdam test sample 0012 using the ConvNeXt-Base checkpoint with seed 42 and TTA. The top row shows the input image, ground truth, prediction, basis-relation graph, and role-coding legend. The lower panels show all 48 learned basis responses grouped post hoc by observed dominant semantic role, together with class-conditioned signatures summarizing basis purity, diversity, usage, and class affinity. The grouping visualizes distributed dictionary use and does not define a fixed class–basis assignment.

(+7.15 points), whereas spectral crosstalk is nearly tied. The complete perturbation definitions, quantitative figure, and per-condition rAUC table are provided in the Supplementary Material; this controlled evaluation follows the broader need to test frequency-sensitive models under distribution shifts [45].

7) Structural Ablation of LDRWA Module: Table XI compares LDRWA with matched replacement blocks on UAVid using the same 512 × 512 FP32 profiling protocol. The baseline reaches 71.27% ± 0.14% mIoU at 11.1 ± 0.4 ms/image. DCNv2, Swin window attention, and standard attention reach

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TABLE XI

MATCHED LDRWA REPLACEMENT COMPARISON ON UAVID UNDER THE SAME 512 × 512 FP32 PROFILING PROTOCOL

Module Configuration	Params (M)	FLOPs (G)	FP32 latency (ms)	mIoU (%)
Baseline (None)	89.6	100.6	11.1±0.4	71.27±0.14
DCNv2 Block	89.6	100.6	11.5±0.2	70.72±0.19
Swin Window Attention	89.6	100.6	11.5±0.2	71.33±0.15
Standard Attention	89.6	100.6	11.2±0.1	71.41±0.12
LDRWA (Ours)	89.7	100.6	12.4±0.5	**71.54±0.17**

70.72% ± 0.19%, 71.33% ± 0.15%, and 71.41% ± 0.12%, respectively. LDRWA reaches 71.54% ± 0.17% with 89.7M parameters, 100.6G FLOPs, and 12.4 ± 0.5 ms/image. These results support the combined use of deformable sampling, local response modulation, and spectral filtering under the evaluated configuration, while the overlapping standard deviations bound the interpretation of small differences among replacement blocks.

F. Visualization Analysis of Core Mechanisms

Extended feature visualizations record the LDRWA effective receptive fields, deformation vectors, and learned complex spectral-filter weights together with the SFO spatial responses, FFT magnitudes, and absolute response differences. The complete visual analysis is provided in the Supplementary Material.

V. CONCLUSION

H-CoReNet combines amplitude-guided channel gating, adaptive spatial receptive-field mixing, shared-dictionary reconstruction, and deformable complex spectral filtering for HRRS segmentation. The region-level and boundary-sensitive results on UAVid, Potsdam, Vaihingen, and LoveDA show the complementary contribution of these operations. The three modules add 0.4M parameters to the backbone-only configuration, while the matched 512 × 512 UAVid profile of the complete ConvNeXt-Base model contains 89.7M parameters and 100.6G FLOPs.

The current architecture increases computational latency on edge devices [36] because it combines fast Fourier transform operations with deformable sampling. Future work will investigate more efficient spectral descriptors, sampling operators, and deployment-aware implementations. These improvements may extend the architecture to multimodal observations [20] and wide-area remote sensing mapping.

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